SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 40%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**

Question Count: 5

SHORT ANSWER TEST

Code of Conduct

I Rakshana A.R (192126002 7) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: R. R

Assessment Tasks

Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

13/10/25 Cloud Computing (CSA1519) CO-1 → AT-3 Short answer test: Rakshana.A.R 192126027

1) Define the Cloud Computing & explain any four characteristics that make it different from traditional on premise computing.

2) Trace the evolution of Cloud computing from hardware evolution to internet software evolution, & show how these changes enabled moder cloud platforms.

3) Explain the role of server virtualization in cloud computing. Different between virtualization as an enabling technology & cloud computing as a service model.

4) Discuss the role of data centres in delivering cloud services...

5) Differentiate Iaas, Paas, Saas, Xaas with 1 suitable real-world eg for each.

1)

Cloud Computing:

- Cloud Computing is a computing technology that provides resources, services, storage, network, software, internet, and database.
- We users just have to pay for their services.

Characteristics:

1) On demand self service:

It obtain resources whenever required without directly intracting with the service provider, using the internet we can access the data.

2) Broad network:

The services can be access through internet using devices such as laptop, monitor...

3) Resource pooling:

This shares resources to multiple user & dynamically allocated according to there demand.

4) Rapid elasticity:

Resources can be quicly increase or decreased according to their workload requirement.

2) # Evolution of cloud computing:

1) **Hardware evolution:**
- Computers were very large & expensive.
- Changes -> increasing processing speed, reducing power consumption lowering the cost of hardware component.

2) **Distributed Computing:**
- Computing tasks are distributed among multiple computers that are connected together through the network.

3) **Virtualization:**
- A single physical server can run multiple virtual machines (VMs).
- Improve hardware utilization & flexibility.

4) **Internet evolution:**
- The speed of the network increases and growth of internet increases, as these are the two important things in cloud computing.

5) **Web & software evolution:**
- web applications & service oriented architectures allowed software to be delivered through browsers instead of being installed locally on a computer memory.

6) Modern cloud computing:

It is a combination that creates platforms that provides services on demand, scalable, remotely accessible resources.

3)

Server virtualization:

- Server virtualization is a process of dividing one physical server into multiple virtual Machines using hypervisor which is basically a software.

- Roles of server virtualization are;
  - * Improve utilization of hard ware components
  - * multiple users can share same hard ware device to store there data separately.
  - * Provides isolation between virtual machines so that the data is safe and stored separately.
  - * Reduce the use of hardware componant.
  - * Drastic decrease in the cost.
  - * Easier to allocate & scale data.

Server virtualization:	Cloud Computing:
◦ Creates virtual resource ◦ Can exist without cloud ◦ Efficient use of hardware ◦ Eg: Multiple VMs running on one server	◦ Deliver resorces to users. ◦ often uses Virtualization. ◦ Resources as service. ◦ Eg: Renting virtual server through AWS

4) **Data center:**

◦ Data center facility containing computing equipments, servers, storage system, power system and cooling systems

◦ This is important because;

◦ **Provides computing resources;**
The servers process application and user requests.

◦ **Provides storage.**
It has large storage systems.

◦ **Provides networking:**
Network equipments are connects cloud to user.

◦ **Ensure availability:**
Backups are stored safely in cloud.

◦ **Provide Scalability:**

It gives scalability along with servers and storage.

- Provides security:

Physical & technical security are strong to protect infrastructure & data.

5.

Models	Name	Function / Meaning	Example
IaaS	Infrastructure as a Service (IaaS)	Provides Virtual Machines, storage, networking.	Amazon EC_2
PaaS	Platform as a service (PaaS)	Development platform, runtime, and tools.	Google App Engine.
SaaS	Software as a service (SaaS)	It provides Ready-to-use software	Google Docs.
XaaS	Anything / Everything as a service	Provides various IT resources delivered as services	AWS Lamba

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